

20.11.18 (E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
Nov-Dec 2018

Max. Marks: 100

Class: SYBTech

Name of the Course: Database Management Systems

Duration: 03 hours
Semester: III

Branch: IT

Course Code: UITC302

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Explain following terms with example a) Hierarchical data model b) Referential integrity c) Multi valued attribute d) Candidate key e) Many to one mapping cardinality	10
Q1(b)	a) Explain different database users b) Explain any five functions of Database administrator.	10
Q 2 (a)	Consider a following information about a Life Insurance Corporation : The Mumbai division office of LIC of India keeps all the necessary information about the policy holders in a database. A policy holder pays a premium until the maturity of the policy or his death, at which time the sum assured and bonus is paid to the nominee. The premium to be paid is worked out based on the age of the person proposed and term of the policy. Mumbai division keeps the following information about each policy holder: SSN, name, address, DOB, description of term policy and annual premium. The corporation has divided its Mumbai division into 15 zones for its convenience. Each zone has its manager. Every zone has number of agents allocated typically ranging from 10 to 20. Every agent must procure a minimum of 10 customers. 1. Draw an EER diagram for the corporation database (state any assumptions you believe you need to make in order to develop a complete diagram). Identify the key attribute (attributes) for each entity and the cardinality of each entity relationship 2. Convert above EER diagram to Relational Model.	10
Q2(b)	What is ECA model? Write syntax for creation of insert and delete trigger with suitable example. OR Explain any five Domain constraint with suitable example (SQL syntax required)	10

Q 3(a)	Explain the steps of query processing in detail. OR What is the need of Normalization in database? Explain 1NF and BCNF with proper example.	10
Q3(b)	What is single level ordered index? Explain primary indexes, clustering index. OR Explain the concept of static and dynamic hashing with suitable example.	10
Q 4(a)	Explain Generalization ,Specialization and Aggregation with example OR Explain the steps of conversion of EER model to Relational Model with suitable example.	10
Q 4(b)	Explain Inner Join, Left Join, Right Join, and Full Join with suitable example.(SQL operations)	10
Q5 (a)	Define deadlock .Explain deadlock detection and Recovery.	10
Q5 (b)	What is serializable schedule? Explain the conditions for the schedule to be view and conflict Serializaable.	10

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End Semester Examinations

Nov – Dec 2018

Max. Marks: 100

Class: SY B.Tech.

Name of the Course: Data Structures

Course Code: UITC303

Duration: 3 hrs

Semester: III

Branch: IT

Instructions:

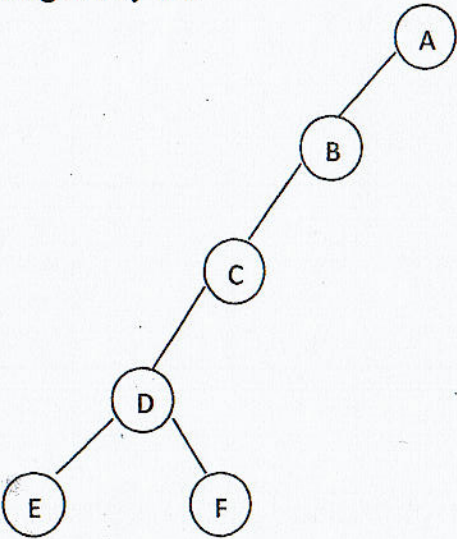
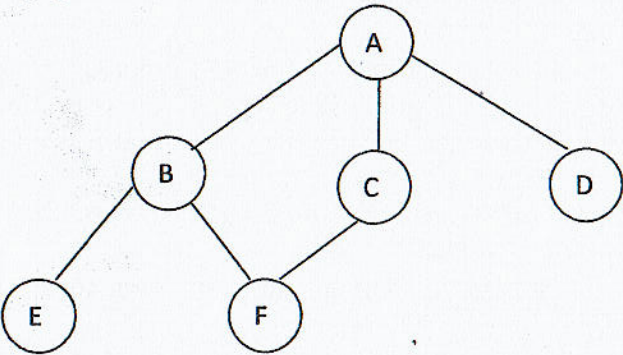
(1) All Questions are Compulsory

(2) Draw neat diagrams

(3) Assume suitable data if necessary

(4) Write programs in 'C' language.

Question No.		Marks
Q 1 (a)	(i) Define Data Structures and explain linear and non linear data structures.	5M
	(ii) Explain Abstract Data Type with an example.	5M
Q 1 (b)	Write a program to implement Stack using an array.	10M
	OR	
Q 1 (b)	Write a program to implement Circular Queue. <i>using an array.</i>	10M
Q2 (a)	(i) Differentiate between singly linked list and doubly linked list.	5M
	(ii) What is a Heap? Explain different types of heaps.	5M
Q2 (b)	Suppose the following list of numbers is inserted in order into an empty Binary Search Tree(BST) 70, 80, 60, 65, 50, 45, 55 (i) Construct the BST (ii) Find inorder, preorder and postorder traversal of BST created (iii) Is the BST constructed an AVL tree? State reasons for your claim. Further if your answer is negative then balance the tree so that it becomes an AVL tree.	10M
Q3 (a)	Write a program to implement linear queue using linked list. (SLL)	10M
	OR	
Q3 (a)	Write a program to implement insertion sort and show the steps to sort the given numbers: 55, 10, 7, 34, 87, 23, 19, 34	10M
Q3 (b)	Using Modulo division method and linear probing, store the values given below in a hash table of size 11. 43, 8, 11, 94, 88, 61, 0, 20, 67, 98	10M
	OR	
Q3 (b)	Write a program to implement binary search.	10M
Q4 (a)	Write an algorithm to convert infix to postfix expression and apply the same to convert following expressions from infix to postfix i) $A / B - C + D * C$ ii) $(A - B) + C / D * E$	10M
	OR	
Q4 (a)	Write an algorithm to perform addition of two polynomials using linked list.	10M

Q4 (b)	(i) Explain AVL tree with an example. (ii) Show the array representation and linked representation for the following binary tree  <pre> graph TD A((A)) --- B((B)) B --- C((C)) C --- D((D)) D --- E((E)) D --- F((F)) </pre>	5M 5M
Q5 (a)	Write a program to create a circular linked list and perform following operations i) Insert into list - <i>Insert-Before</i> ii) Delete from list - <i>Delete-After</i> iii) Display the list OR	10M
Q5 (a)	Write a program to reverse a singly linked list.	10M
Q5 (b)	(i) Apply Breadth First Search and Depth First Search traversal on the below graph :  <pre> graph TD A((A)) --- B((B)) A --- C((C)) A --- D((D)) B --- E((E)) B --- F((F)) C --- F </pre> (ii) Explain how graphs can be represented in memory with the help of a suitable example.	5M 5M

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End Semester Exam
Nov-Dec 2018

Max. Marks: 100

Class: SY-BTech

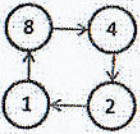
Duration: 3 hoursName of the Course: **DLDA** (Digital Logic Design and Applications)

Semester: III

Course Code: UITC304

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams if necessary
- (3) Assume suitable data if necessary

Q No		Max Marks
Q1	Do the following number conversions (show the calculations clearly):- (a) $(9A)_{16} = (?)_{BCD}$ (d) $(0.01)_{10} = (?)_{16}$ (b) $(9A)_{16} = (?)_4$ (e) $(?)_{10} = (FF)_{Gray\ code}$ (c) $(9A)_{16} = (?)_3$	15
Q2(a)	Implement the function $Y = \sum m(4, 6, 7, 12, 15)$ using a single 8:1 multiplexer. ----- OR ----- Explain the concept of Map Entered Variable (MEV) with a suitable example.	10
Q2(b)	Draw the circuit diagram (using NAND gates) for a T flip flop with asynchronous preset and clear. ----- OR ----- Explain the term "race condition" in a JK flip flop, and how a master slave flip flop is used to avoid the same.	10
Q3(a)	 (i) Using JK flip-flop in toggle mode (as a T-flip-flop), design the sequence generator shown in this diagram which repeats the pattern 8-4-2-1.... (ii) Draw the complete state diagram which shows the unused states also.	15
Q3(b)	Draw the complete state diagram (including unused states) for the counter shown in Fig2 (Page 2 below). Assume that the initial state is 1111.	15
P.T.O.		

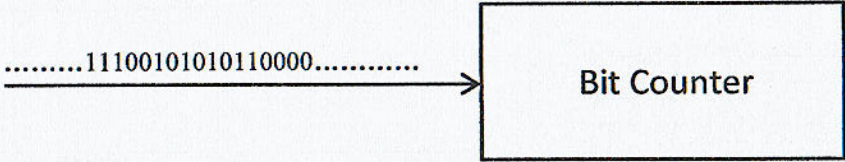
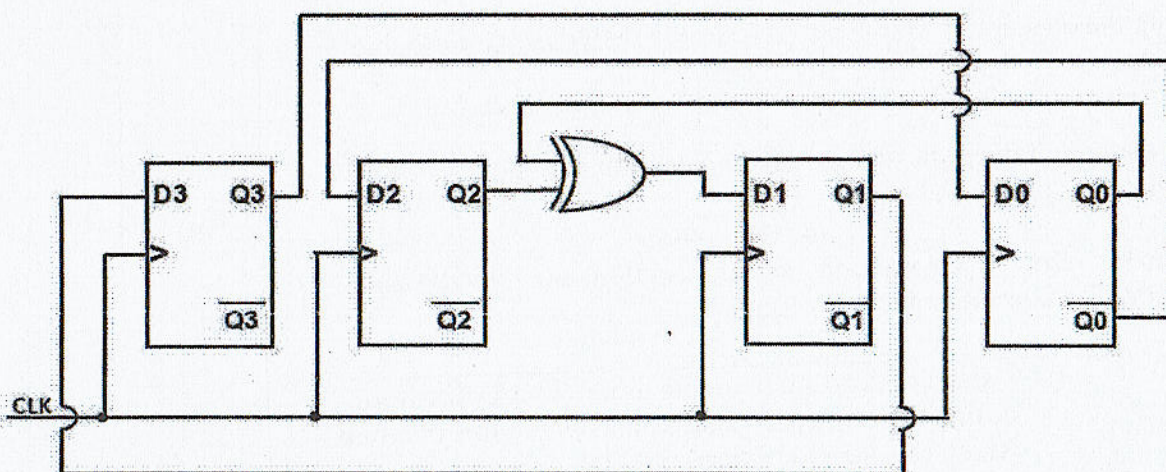
Q4	(a) Explain the need for and advantages of using VHDL for designing digital circuits. (b) Write the behavioral VHDL model for a 3:8 decoder. ----- OR ----- (c) Explain the data type "STD_LOGIC" in VHDL. (d) Write the structural VHDL model for a two input EXOR gate which uses only NOR gates.	20
Q5	11100101010110000.....  A continuous serial stream of bits is sent to a "Bit Counter", which counts the number of "0" bits in the incoming data. Without actually designing the counter, describe and/or draw in as much detail as possible, what the internal circuit will look like. Assume that the bit stream and the counter share the same clock. Make other suitable assumptions if necessary.	15

Fig2:



END

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K. J. Somaiya College of Engineering, Mumbai-77

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End Semester Exam

Nov – Dec 2018

Max. Marks:100

Class: SYBTech

Name of the Course: Theory of Computers

Course Code: UITC305

Duration: 03 Hrs

Semester: III

Branch: IT

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Define Automata. Design a DFA over $\Sigma = \{a, b, c\}$ accepting strings such that every two consecutive a's are followed by 'bc' and every 2 consecutive c's are followed by ba. Give Formal definition of automaton along with processing over two strings being accepted. OR Define NFA. Design a DFA over $\Sigma = \{a, b, c\}$ that accepts strings such that i. If starting with 'a' strings are of minimum length 5 ii. If starting with 'b' string are of minimum length 4 and ends with "cb" Give Formal definition of automaton along with processing over two strings being accepted	12
Q 1 (b)	Construct an ϵ-NFA that accepts L. $L = \{ \text{run, runs, arun, barun, varun} \}$ Convert the same into equivalent DFA. Write tuples for both the automata's.	08
Q2 (a)	How DFA and NFA are equivalent? How it can be proved? Explain with a suitable example.	08
Q2 (b)	Construct an ϵ-NFA that accepts strings over $\Sigma = \{a, b, c\}$ i. Starting with ab and ending with cb OR ii. Starting with bc and ending with ca OR iii. Starting with ca having substring bb and ending with ac or cb Convert the same into equivalent DFA. Minimize the derived DFA with state elimination method.	12
Q3	Is it not possible to construct a DFA/ NFA to accept language of even length palindromes? Justify your answer with necessary proof. (Consider any alphabet with two symbols in the required example). OR Define Regular Expressions. With suitable example give rules of converting Regular expression to Finite Automata and vice versa.	10
Q4 (a)	Write in detail steps of converting grammar in Chomsky Normal Form into equivalent Greibach Normal Form. Explain every step with a suitable example.	08

Q4 (b)	Convert the given CFG in CNF and GNF $S \rightarrow A B$ $A \rightarrow aBa \epsilon$ $B \rightarrow bAb \epsilon$	12
Q4 (b)(c)	Consider the language $L = \{a^n b^m c^{n+m} \mid m, n \geq 1\}$. Construct PDA M that accepts the given language. Also construct equivalent CFG for L such that $L(G) = L(M)$ OR Construct a PDA that accepts the language of even A's and even B's. Show processing over any acceptable string (length ≥ 8) using turnstile notation.	10
Q5 (a)	Design a Turing Machine that accepts the language over $\Sigma = \{a, b\}$, such that sub-string "bab" is not present in the strings.	10
Q5 (b)	Explain components of Turing Machine with a suitable diagram. Explain how Turing machines can be defined formally with a suitable example. OR Explain Church- Turing Thesis. With suitable diagrams explain variants of Turing Machine.	10

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